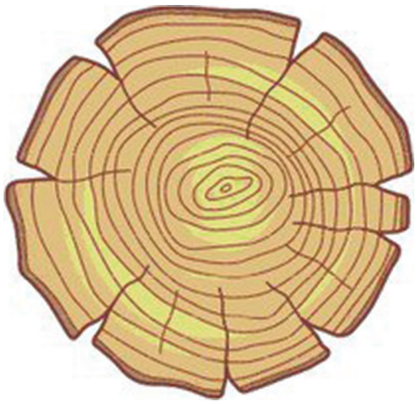
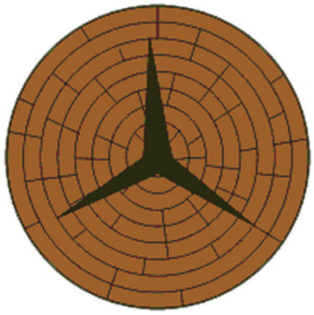
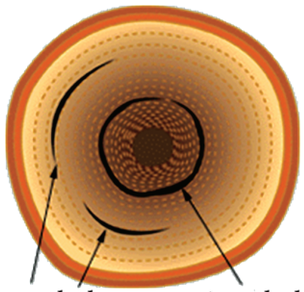


2. **Heart Shakes:** These cracks occur at the centre of the cross section of the tree and it extends from pith to sapwood in the direction of medullary rays. These cracks occur due to shrinkage of interior part of tree which is approaching maturity. It divides the cross-section of the tree into two or four parts.



3. **Cup Shakes**: Cup shakes separate the whole or part of one annual ring from another and are caused by wind and frost in the growing tree.



4. **Radial Shake:** These are similar to star shakes and occur in felled timber when exposed to the sun during seasoning. Radial shakes are generally irregular, fine and numerous.

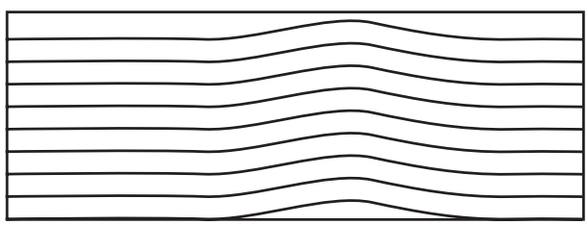


5. **Rind Galls:**



A peculiar curved swelling found on the body of the tree is known as rind galls. They develop at points from where branches are improperly cut-off or removed.

6. **Upsets:** Upsets are portions of the timber in which the fibres have been injured by crushing, bending or shocks during growth of the tree.



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